Call for AI Development Assistant

Course: Quantum Field Theory 1

Physics Department

Instructor: A.Ardehali

Overview

The AI Development Assistant will support the instructor in building, deploying, and refining an agentic workflow for implementation and automated verification of graduate student projects, as well as in refining a chatbot (already built by the instructor using BotPress) for collecting and grading the student homeworks. The assistant will also provide technical coaching to students on tools essential for the course (VPN, GitHub, Overleaf, Mathematica, etc.) through four scheduled 30-minute sessions during six selected weeks across the semester.

Total Workload: 5–7 hours/week.

- 2 hours/week with the course instructor (planning, review, integration).
- 3 hours/week on independent development and documentation
- 2 hours student support (four slots, each 30 minutes), in six rounds.

Required Skills

- Familiarity with Python and GitHub (repositories, GitHub Actions).
- Proficiency in LLM-based coding assistants (e.g., GapCode, Claude Code).
- English proficiency; ability to write clear technical documentation/user guides.
- Ability to help with VPN/Git/Overleaf/Mathematica installation and setup.
- Patience.

Preferred Additional Skills

- Background in physics (especially: path-integrals in quantum mechanics).
- Experience as a teaching assistant.

Timeline

Weeks 1–3: Infrastructure & ChatBot. Weeks 4–6: Scaffolding & Student Preparation. Weeks 7–10: Iterative Support & Refinement. Weeks 11–12: Finalization & Handover